Personal Statement

As a driven and ambitious student pursuing degrees in Computer Science, Electrical Engineering, and Mechanical Engineering, my academic journey has been marked by a relentless pursuit of excellence and a deep passion for innovation. With a diverse set of experiences and accomplishments, I have acquired valuable skills and knowledge that have shaped my current interests and solidified my vision for the future. As I reflect upon my past activities and achievements, it becomes evident that my unwavering commitment to advancing the engineering field makes me a deserving candidate for grant consideration.

My past activities have not only honed my technical prowess in the engineering disciplines but also fostered my capacity to adapt and integrate knowledge from various fields. Leading interdisciplinary teams in projects such as the ICE Go Kart, MOCAP v2, and the Vehicular Control System, I have demonstrated my ability to leverage the synergies between computer science, electrical engineering, and mechanical engineering to create groundbreaking solutions. My voluntary work on the Faculty Research Database and the ASME Tiger Burn event further showcases my dedication to fostering a collaborative environment and promoting innovation within the engineering community.

In the realm of research, I have delved into projects like the "LPBFAM Distortion Prediction" and "Machine Learning on FPGAs for Power Electronics Digital Twin Systems". These experiences have nurtured my interests in additive manufacturing, simulation, machine learning, and their applications in electrical and mechanical systems, while also exposing me to the vast potential of computer science in engineering applications. As a testament to my commitment to pushing the boundaries of knowledge and innovation, I have been awarded grants and scholarships such as the Pi Tau Sigma Research Grant and the Magellan Scholar Grant.

Currently, I am passionate about exploring the confluence of artificial intelligence, advanced manufacturing techniques, and computer science to create more sustainable, efficient, and innovative systems. Moreover, I am focused on enhancing my understanding of computer architecture, low-level programming, and their applications in creating unique and challenging projects, such as the "Tic Tac Toe Game for the Gameboy Advanced Console".

As I look towards the future, I aspire to contribute significantly to the engineering field by pursuing a PhD. I believe that continued education will enable me to unlock the untapped potential of emerging technologies and their applications in various industries. I envision a future where I lead interdisciplinary teams in developing novel solutions to pressing global challenges, such as climate change, resource scarcity, and energy efficiency. Additionally, I hope to mentor and inspire the next generation of engineers and advocate for increased diversity and inclusion within the field.